FIT2179 Data Visualisation

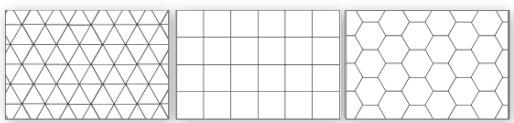
Course Notes

BIN MAPS

12 September 2026

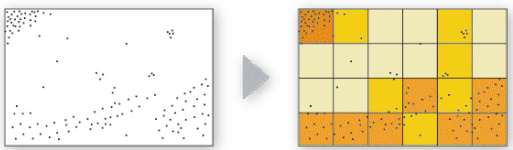
BIN MAPS

Bin maps are related to choropleth maps, but they use a regular tessellation instead of irregularly shaped areas. The tessellation can consist of triangles, squares, or hexagons. A single tessellation unit is called a bin, and each bin is colour-coded by an attribute value, just as in a choropleth map.



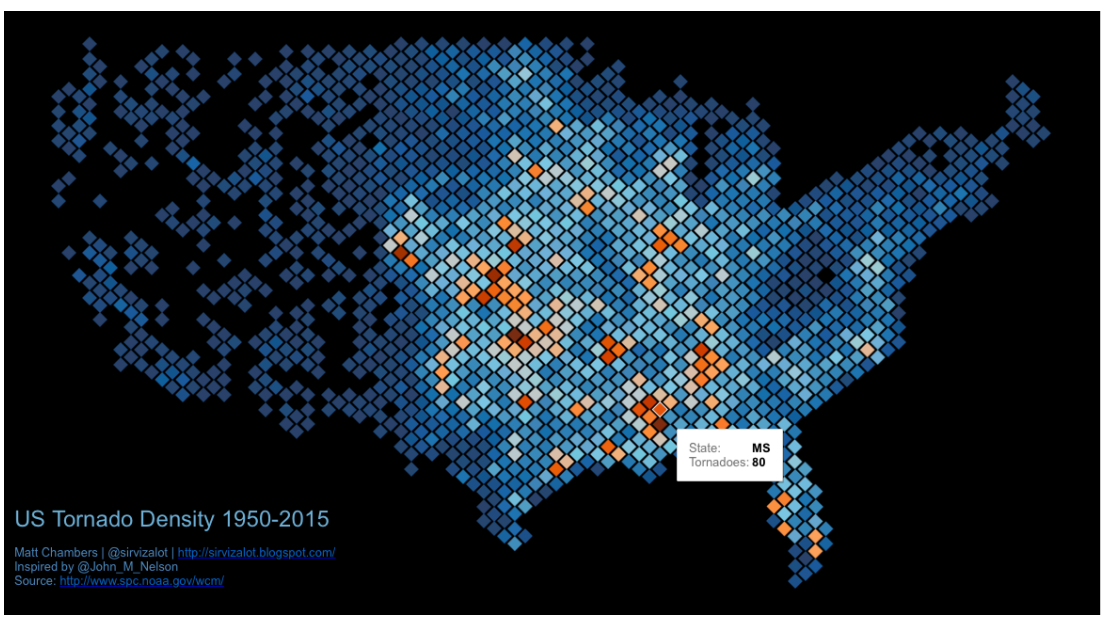
CONSTRUCTING A BIN MAP

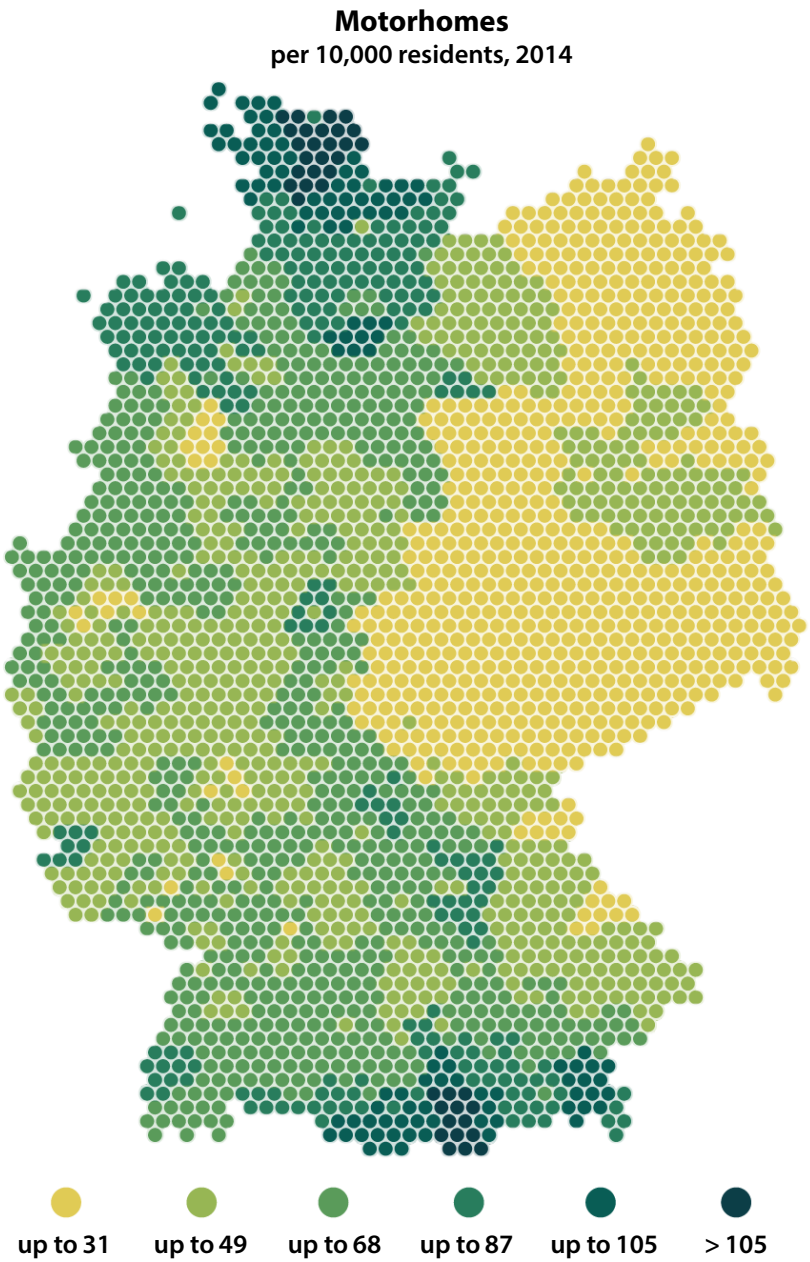
A quantitative bin map is usually built from point geometry, for example, a set of points of sightings of a rare bird species. A tessellation type and size are selected, then the number of sightings falling inside each bin is counted. This derived attribute is mapped to a colour, which is used to draw the bin.



Lines and areas can also be used to build bin maps. One can count the number of lines that intersect a bin area. The tornado map below is an example built from tornado tracks (line geometry).

As with quantitative choropleth maps, darker colours usually represent higher values. This can be inverted when the background is dark, as in the tornado map.





Map from
Stolz, M. & Block, J. Deutschlandkarte, 2012, Zeit Magazin

STANDARDISED BIN MAPS

Bin maps have one advantage over choropleth maps: because all bin areas are the same size, standardising by area is not necessary. Standardisation is still needed, however, when the mapped value is a fraction of a total.

The example map shows the distribution of motorhomes in Germany in 2014. If it showed raw totals of camper vans, it would largely reproduce population density since areas with more people also have more motorhomes. Instead, the number of motorhomes in each bin is divided by the number of people living in that bin. The resulting rate reveals an interesting spatial pattern: motorhomes are scarce in eastern Germany and concentrated in the far north and far south.

SIZE CHANNEL

Varying the size channel in a bin map is rare, but still effective.



Map from
Bertin, J. 1983, Semiology of Graphics. University of Wisconsin